MODULE SYLLABUS

Network Architectures

MODULE CODE	CCS1320
MODULE TITLE	Network Architectures
PROGRAMME	BSc (Hons) in Computer Science
DEPARTMENT	Computer Science
CREDITS	10
STAGE OF STUDY	BSc
SEMESTER/SESSION	Spring 2026
RE-ASSESSABLE	Yes
COMPENSATABLE	Yes
LOCATION	Thessaloniki
STAFF	Dr. Georgios Kalfas
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ACCREDITATION	The programme is accredited by the British Computer Society (BCS)

DESCRIPTION

The emergence of networked computers as the computing platform for many applications affects every aspect of the study and use of computing. This module introduces modern communication systems. It offers a comprehensive overview of the fundamental principles, theories, and practices employed in computer networks and aims to develop a foundation that can be used as the basis for further study or research in the field.

AIMS

This module aims to:

A1 examine the structure and architecture of data communication systems and computer networks
A2 discuss hardware components required to form computer networks
A3 describe challenges for achieving effective and efficient data communications and their solutions
A4 investigate various networking and internetworking protocols and standards
A5 explain the layered approach of the TCP/IP protocol stack

LEARNING OUTCOMES

By the end of the module, a student will be able to:		Link to aims
LO1	define basic computer network technology concepts	A1, A2
LO2	explain principles of data communications systems and their components	A1, A2
LO3	distinguish networking devices and components and explain their operation	A1, A2
LO4	describe working principles behind key network technologies, standards, and protocols	A3, A4, A5
LO5	explain and compare flow control and error control mechanisms	A3, A4
LO6	apply error detection techniques used in computer communications	A3, A4
LO7	explain, analyse and compare key networking protocols and their components	A4, A5

HOW DOES THIS MODULE FITS INTO THE CURRICULUM?

Computer networks are an integral part of modern information and communication technologies infrastructure. The different types of computer networks have a variety of specifications and requirements that are supported by different network architectures. Therefore, knowledge of architectural networks is very useful for the next modules of the programme, as well as a valuable asset for current and future computer science professionals.

TEACHING & LEARNING METHODS	Total Contact Hours: 24

The following teaching & learning methods will be employed:
The unit is delivered through formal lectures (20 hours), a consolidation session (2 hours) and a revision session (2 hours). A diverse number of fun, interactive learning activities are planned to promote and facilitate active learning. These activities encourage questions and classroom discussions, and enable students to explore, reflect on, and apply what they have learned. Examples of such activities include group discussions and competitions, utilization of interactive multimedia material in the classroom, exploitation of virtualizations to support the realization of concepts, gamified elements to strengthen engagement, etc.

ASSESSMENT METHODS

#	Students will be assessed by:	Submission Week	% contribution	LOs assessed
1	Quiz	8	35%	LO1, LO2, LO3, LO4
2	Final Exam	RA13	65%	LO1 - LO7

FEEDBACK PROVISION

The following methods will be used to provide feedback to students:

- Formative feedback: Formative assessment of the students and feedback is provided via the interactive activities that take place during contact hours.
- Summative feedback: Following summative assessment verbal feedback sessions take place during the contact hour that follows the assessment to explain to the students the model answers and common misconceptions. Feedback for the Final Exam will be provided via written remarks on the exam papers/feedback forms.

The feedback handbook, available here, aims to give you a better understanding of feedback; what it is for and how to use it.

ACCESS TO MODULE MATERIAL (Notes, handouts, announcements etc.)

All material used in this module's classes are available in electronic form through Google Classroom with class code **5rflkkmu**.

RECOMMENDED TEXTBOOK(S)

Kurose James, Ross Keith, Computer Networking: A Top-Down Approach, latest edition, Pearson

LIST OF REFERENCES / ADDITIONAL RECOMMENDED READING

Comer Douglas, “Computer Networks and Internets”, latest edition, Pearson
Forouzan Behrouz A., Data Communications and Networking, latest edition, McGraw Hill
Stallings William, Data and Computer Communications, latest edition, Prentice Hall
Tanenbaum Andrew, Computer Networks, latest edition, Prentice Hall
Forouzan Behrouz, TCP/IP Protocol Suite, latest edition, McGraw Hill

OUTLINE	
WEEK/ SESSION	
#1	Introduction – Basic Networking Concepts
#2	Layered Approach of Network Models
#3	Network Applications Fundamentals
#4	Modern Network Applications
#5	Local Area Networks – Fundamentals and Devices
#6	Consolidation week
#7	Error Detection - Flow and Error Control
	Easter Break
#8	Wide Area Networking
#9	Internetworking I
#10	Internetworking II
#11	Internetworking III
#12	Revision week

EMPLOYABILITY PROFILE

This module contributes to your employability profile by enhancing the following Graduate Attributes: